



Causal Inference for Spatial Cross-Sectional Data: A Geographical Convergent Cross Mapping Method Accounting for Spatial Autocorrelation

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Abstract

Causal inference for spatial cross-sectional data is an essential approach to reveal the driving mechanisms of geographical phenomena. As an emerging method for causal inference with spatial cross-sectional data, Geographical Convergent Cross Mapping (GCCM) is easily disturbed by “false neighbors” caused by spatial autocorrelation, and this can lead to misjudgment of causal direction or overestimation of causal strength. In view of this problem, this paper proposes a Geographical Convergent Cross Mapping method accounting for Spatial Autocorrelation (SA-GCCM). The method introduces spatial path roughness to measure the smoothness of attribute value changes between two points, and constructs an iterative neighbor screening mechanism to eliminate the “false neighbors” problem caused by spatial autocorrelation. In addition, the proposed method identifies true causal relationships by examining how the predictive performance changes with the spatial path roughness. To verify the effectiveness of the SA-GCCM method, this study conducted experiments on two typical datasets with clear causal relationships: nighttime-light data and topographic factors, and net primary productivity and climatic factors. The experimental results show that SA-GCCM can effectively address the issues of causal direction misjudgment and causal strength overestimation in the original GCCM method. This study not only improves the robustness of the GCCM method, but also provides methodological support for handling geographical data with spatial autocorrelation.

Keywords Geographical Convergent Cross Mapping · Causal Inference · Spatial Autocorrelation · Spatial Path Roughness

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Introduction

The occurrence and evolution of geographical phenomena are governed by specific driving factors, thus forming a geographical space where spatial correlation and spatial heterogeneity coexist. For a long time, correlation analysis has been the mainstream method for exploring the relationships between geographical variables (Brunsdon et al. 1998; Rahmawati et al. 2025; Zhao et al. 2026). However, relying only on traditional correlation analysis can hardly reveal the formation mechanisms and evolutionary laws of complex geographical phenomena, and may even lead to spurious causality (Su et al. 2023). Therefore, causal analysis plays an important role in geographical research and has received increasing attention.

Traditional causal inference mainly relies on statistical methods, including the Granger causality test (Granger 1969; Chen et al. 2004) and various regression models.

Nevertheless, methods based on linear assumptions and stochastic processes often fail to capture the nonlinear dynamic characteristics of complex systems and are easily affected by spurious correlations. To overcome this limitation, Sugihara et al. (2012) proposed the Convergent Cross Mapping (CCM) method based on Takens' embedding theorem (Takens 2006) and state-space reconstruction theory (Deyle and Sugihara 2011). The core logic of this method is: if variable X is a driving factor of variable Y , the dynamic information of X will be encoded in the evolutionary history of Y , so that X can be predicted using the reconstructed manifold of Y .

Taking the watershed precipitation-runoff system as an example, let X denote daily precipitation and Y denote daily runoff. The true causal relationship is that precipitation drives runoff changes ($X \rightarrow Y$), while the feedback effect of runoff on regional precipitation is negligible at small scales. The CCM procedure is summarized in three steps: First, high-dimensional embedding vectors are constructed using the time series of Y and its time-lagged terms to form the shadow manifold M_Y for Y . Next, the K -nearest neighbors of the point at the target time are identified in M_Y . These neighbors are then mapped to the time series of X , and the predicted value of X at that time is calculated by weighted average. Finally, the library size for manifold construction is gradually increased, and the Pearson correlation coefficient ρ between predicted and observed values is computed. If ρ increases significantly and converges to a stable value as the library size increases, this confirms the existence of a true causal relationship from X to Y .

Although CCM has prominent advantages in analyzing nonlinear dynamic systems, its traditional form and subsequent extensions (Clark et al. 2015; Brunton et al. 2016; Coufal et al. 2017; Leng et al. 2020; Díaz et al. 2022; Martin et al. 2024) were primarily designed for time-series data. Applying these methods directly to spatial cross-sectional data is highly challenging. Unlike time series data, spatial cross-sectional data are simultaneous observations at different locations and lack the natural temporal order of "cause preceding effect". If the time span is insufficient to reflect the changes of geographical variables, it will be difficult to detect causal relationships between them. For example, latitude zonation leads to spatial distribution differences in global soil types, a process that may take a very long time to show identifiable changes. Temporal fluctuation is one perspective for causal inference, and spatial heterogeneity is another (Gao et al. 2022).

To extend the idea of CCM to spatial cross-sectional data, Gao et al. (2023) proposed the Geographical Convergent Cross Mapping (GCCM). GCCM replaces time lag with spatial lag and constructs a shadow manifold to simulate the system state space for cross-mapping prediction. Theoretically, GCCM can not only reveal potential causal

relationships, but also identify patterns like unidirectional driving and bidirectional feedback through convergence analysis. Later, Zhang et al. (2025) extended GCCM and proposed the Geographical Pattern Causality (GPC) model, which can determine whether an independent variable promotes or inhibits a dependent variable.

As a breakthrough extension of convergent cross mapping theory in the spatial domain, GCCM successfully solves the problem of causal inference for spatial cross-sectional data. As a new tool in earth system science, ecology, climatology, and other fields, GCCM has been widely used in studies on the driving mechanisms of ecological environment quality (Liang et al. 2024; Shi et al. 2025), the influencing factors of surface albedo (Yuan et al. 2024), and the relationship between habitat quality and landscape pattern (Zhang et al., 2025a). It provides strong support for revealing the nonlinear causal mechanisms of complex systems.

Despite its theoretical expansion, GCCM is still limited by the spatial autocorrelation effect of geographical variables in practical applications. The core principle of GCCM is to find nearest neighbors in the state space to recover the dynamic characteristics of the system. However, the inherent spatial autocorrelation of geographical variables often makes the neighbors in the variable state space only spatially adjacent points, not points with similar dynamic properties. This phenomenon reduces manifold-based cross prediction to spatial neighbor-based interpolation, causing causal inference biases. Existing studies have shown that GCCM produces inference biases in several typical application scenarios, such as approximate bidirectional causal strengths, reversed causal directions, or even failure to detect obvious unidirectional driving relationships (Liang et al. 2024; Yuan et al. 2024; Tian et al. 2024; Han et al. 2025; Shi et al. 2025; Wang et al. 2025; Zhang et al., 2025b).

To address the "false neighbors" problem caused by spatial autocorrelation, this study proposes a Geographical Convergent Cross Mapping accounting for Spatial Autocorrelation (SA-GCCM). This method optimizes the neighbor screening mechanism by removing neighbors that only show state similarity due to spatial autocorrelation, to improve the reliability of causal inference. The core idea of SA-GCCM is to introduce the spatial path roughness to measure the smoothness of attribute value changes between two points in geographical space, to judge whether the state similarity between two points comes from spatial smoothness or the internal dynamic structure of the system. Specifically, during the K -nearest neighbor search, we test the path characteristics between candidate points and the target point, and eliminate redundant points with roughness below the threshold to identify valid neighbors that are geographically distant but truly similar in dynamic state. In addition, this study identifies true causal relationships by analyzing

how cross-prediction performance changes with the spatial path roughness.

The rest of this paper is organized as follows. Section 2 describes the SA-GCCM method in detail. Section 3 verifies the proposed method using two typical cases in geographical research. Section 4 discusses the SA-GCCM method. Section 5 concludes the paper.

Methodology

The overall framework of the proposed method is illustrated in Fig. 1, which consists of three main steps. First, the original data are detrended and standardized. Next, high-dimensional embedding vectors are constructed using spatial lags to reconstruct the variables' shadow manifold, thus recovering the system's dynamics. Second, a spatial path roughness index is introduced to measure the smoothness of attribute changes between two points in geographic space. Based on this index, a dual-constrained iterative screening mechanism is designed for nearest neighbors, thus yielding a valid set of neighbors with truly similar states. Third, cross-mapping prediction is performed using the selected neighbors, and its prediction accuracy is evaluated. Finally, true causal relationships are identified based on the characteristic that prediction performance remains stable as the roughness threshold increases.

GCCM Model

Unlike CCM which reconstructs the state space with time lags, GCCM innovatively introduces the concept of spatial lags and successfully applies CCM theory to spatial cross-sectional data. The basic idea is: if variable X is a driving factor of variable Y , the dynamic information of X is encoded in Y 's state space. Therefore, Y 's shadow manifold can theoretically be used to predict X 's state. Specifically, the algorithm searches for multiple neighbors of the target point on Y 's manifold and maps them to X 's manifold. The weighted average of these neighbors' corresponding X values is calculated through simplex projection to realize X 's prediction.

For variables X and Y covering N spatial units in the study area, their state spaces are reconstructed by spatial lags. For any spatial unit s , its embedding vector $\psi(x, s)$ is defined as:

$$\psi(x, s) = \langle h_s(x), h_{s(1)}(x), \dots, h_{s(E-1)}(x) \rangle \quad (1)$$

$$\psi(y, s) = \langle h_s(y), h_{s(1)}(y), \dots, h_{s(E-1)}(y) \rangle \quad (2)$$

where E denotes the embedding dimension, and $h_{s(i)}(x)$ represents the i -th order spatial lag observation for spatial unit s . By iterating through all spatial units, the shadow manifolds M_X and M_Y for variables X and Y are constructed, respectively:

$$M_X = \{\psi(x, s_1), \dots, \psi(x, s_N)\} \quad (3)$$

$$M_Y = \{\psi(y, s_1), \dots, \psi(y, s_N)\} \quad (4)$$

For the target spatial unit s , the K -nearest neighbors $\{n_1, \dots, n_K\}$ of its state vector $\psi(y, s)$ are identified within M_Y . Simplex projection is then used to make predictions using the X observations of these neighbors:

$$\hat{X}(s) | M_Y = \sum_{k=1}^K w_k X(n_k) \quad (5)$$

where the weight w_k is calculated based on the exponential decay of the state-space distance. Finally, the Pearson correlation coefficient $\rho_{X \rightarrow Y}$ between the predicted value $\hat{X}$ and the observed X is computed. According to the convergence criterion, if $\rho_{X \rightarrow Y}$ increases significantly and tends to stabilize as the library size grows, a causal relationship from X to Y is confirmed.

The False Neighbor Problem Caused by Spatial Autocorrelation

Geographical variables exhibit strong spatial autocorrelation, and their spatial distribution often shows a smooth surface at the local scale. As a result, the nearest neighbors in state space identified by the GCCM algorithm are actually neighboring points in geographic space. For any spatial point s , the point s' in its spatial neighborhood has a value similar to that of s . This causes the distance $d(\psi(y, s), \psi(y, s'))$ between their embedding vectors $\psi(y, s)$ and $\psi(y, s')$ in the E -dimensional space to be close to zero. In this case, when searching for the K -nearest neighbors of $\psi(y, s)$, GCCM tends to find points that are geographically close to s . In this scenario, the topological structure of the reconstructed manifold is no longer governed by the underlying dynamics, but instead dominated by geographic distance. GCCM uses the simplex projection algorithm for cross-prediction, which essentially carries out a weighted estimation using the X values of the K -nearest neighbors on M_Y . When the above situation occurs, the neighbor set $\{n_k\}$ essentially becomes the geographic neighbors of s .

Under such conditions, the original dynamic prediction $\hat{X}(s) | M_Y$ based on manifold mapping mathematically

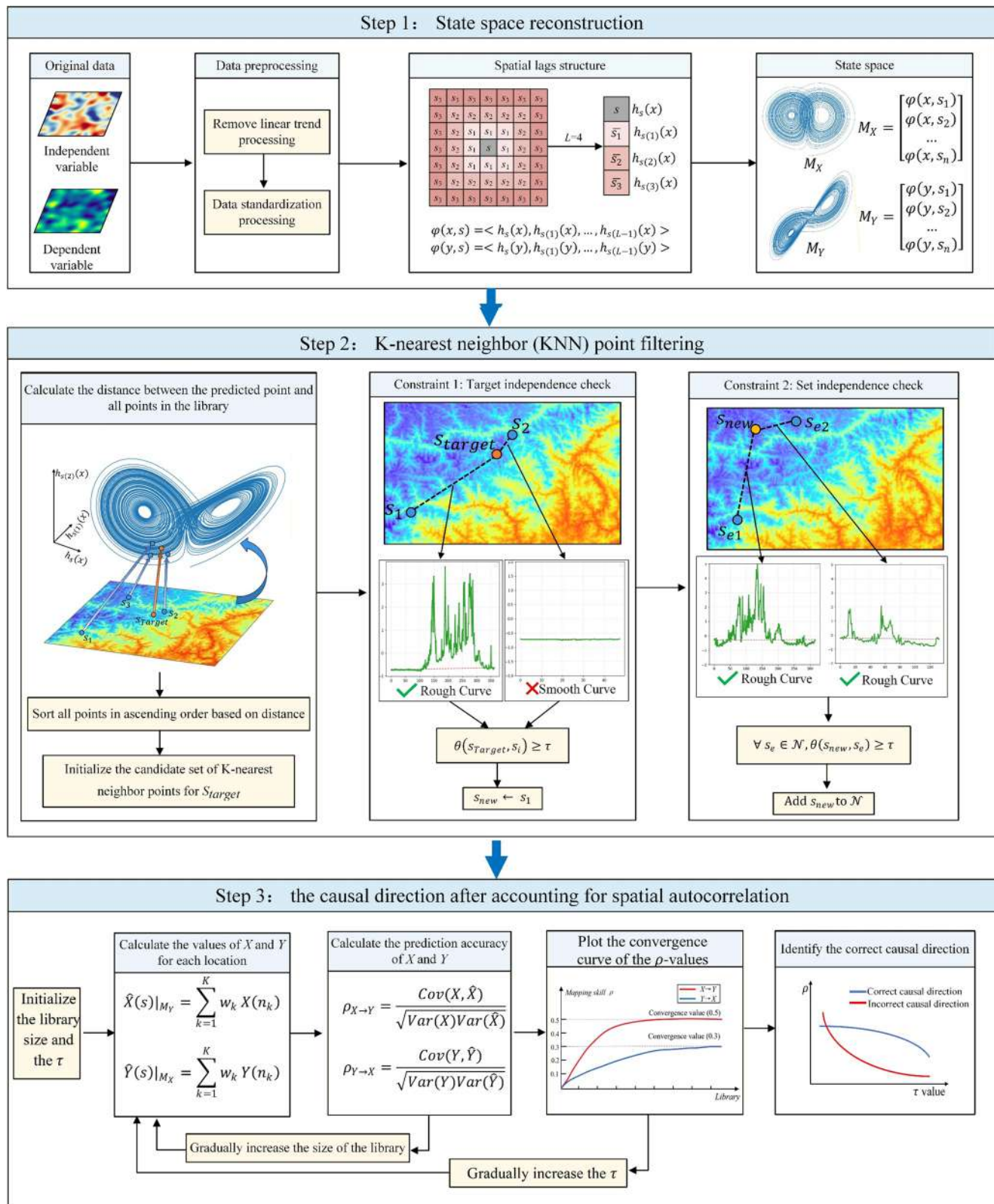


Fig. 1 Schematic workflow of the SA-GCCM method for spatial causal inference. Schematic overview of the SA-GCCM method

reduces to inverse distance weighted interpolation. If X also has strong spatial autocorrelation, interpolation using spatially adjacent points will inevitably yield high prediction accuracy. This “false neighbor” effect caused by spatial structure hides the real dynamic coupling between variables. It leads to the false judgment that Y has a strong causal effect on X , and this judgment is usually opposite to the actual causal direction.

To obtain reliable causal inference results, it is necessary to eliminate “false neighbors” that only show state similarity because of short geographic distance in the K -nearest neighbor screening stage, and retain valid neighbors that are geographically distant but truly similar in dynamic state.

SA-GCCM Model

Spatial Path Roughness Measurement

To eliminate the “false neighbors” caused by spatial autocorrelation, this study defines spatial path roughness to judge whether the similarity between two points in geographic space comes from spatial autocorrelation. For any two points in geographic space, namely the target point s_{target} and the candidate neighbor s_{cand} . Prior to the K -nearest neighbor screening step, it is ensured that the s_{target} and s_{cand} exhibit high similarity in the state space. First, construct a straight path P between them. We sample m points at equal intervals along path P to get the observation sequence $V_{\text{obs}} = \{v_1, v_2, \dots, v_m\}$, where v_i is the observed value of variable Y at the i -th sampling point. Meanwhile, we define the linear trend line connecting the initial value v_1 and the final value v_m as the ideal smooth path, and its sequence $V_{\text{ideal}} = \{l_1, l_2, \dots, l_m\}$ satisfies:

$$l_i = v_1 + (v_m - v_1) \cdot \frac{i-1}{m-1}, i = 1, \dots, m \quad (6)$$

This linear trend represents the smoothness assumption under the only influence of spatial distance. The spatial path roughness is defined as the root mean square error (RMSE) between the observed path and the ideal linear trend:

$$\theta(s_{\text{target}}, s_{\text{cand}}) = \sqrt{\frac{1}{m} \sum_{k \in \Omega} (v_k - l_k)^2} \quad (7)$$

where Ω denotes the set of observations along the path.

The physical meaning of this index is as follows: When θ is small (close to 0), the attribute values change smoothly from s_{target} to s_{cand} , which conforms to the characteristics of spatial autocorrelation. In this case, the state similarity between the two points is most likely caused by spatial autocorrelation (Corresponding to the smooth curve in Fig. 2). When θ is large, it indicates obvious fluctuations in attribute values, severe topographic fragmentation or strong spatial heterogeneity between the two points. If the two points are still close in the state space under such conditions, such similarity transcends geographic barriers and is more likely derived from the dynamic attractor structure inside the system (Corresponding to the rough curve in Fig. 2).

K-nearest Neighbor Screening

Based on the above spatial path roughness measurement, this study proposes an iterative neighbor screening algorithm to replace the traditional screening method based only on Euclidean distance in GCCM. Two critical constraints are added in this algorithm.

Constraint 1 (Target Independence Constraint): This constraint evaluates the path roughness between a candidate point q_i and the target point s_{target} . If $\theta(s_{\text{target}}, q_i) < \tau$, it indicates that the state similarity between the two points is highly likely caused by geographic proximity. Thus, q_i is classified as a “false neighbor” and directly discarded.

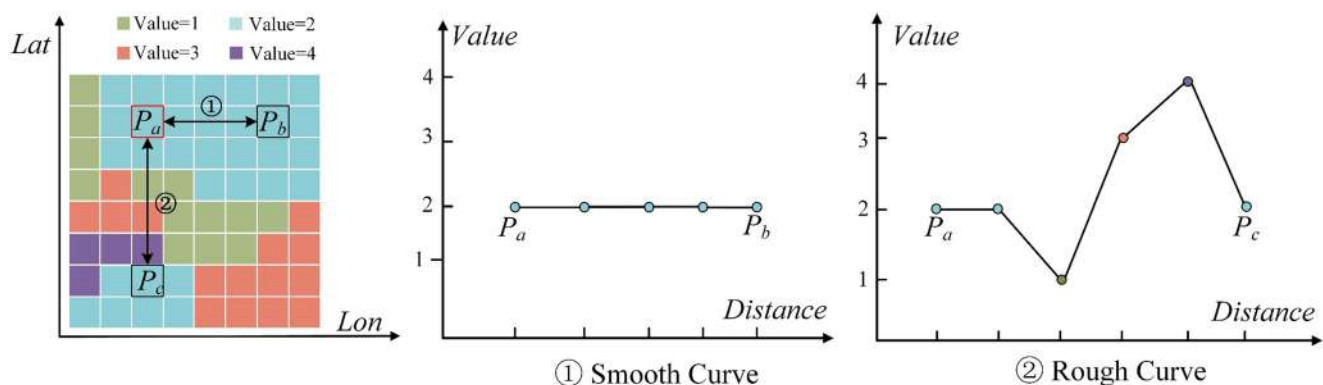


Fig. 2 illustrates the principle of “false neighbor” detection. Path ① represents a smooth sequence induced by spatial autocorrelation, where variable values change little with increasing distance. Thus, P_b

is a “false neighbor” point. Path ② represents a rough sequence, and P_c is a true neighbor point. Schematic diagram of “false neighbors” detection

$$\text{Reject } q_i, \text{ if } \theta(s_{\text{target}}, q_i) < \tau \quad (8)$$

Constraint 2 (Set Independence Constraint): To prevent the selected K -nearest neighbors from clustering in the same geographic area, it is necessary to ensure that the newly added candidate q_i satisfies Constraint 1 with every existing neighbor s_{exist} in the current set $\mathcal{N}$.

$$\text{Reject } q_i, \text{ if } \exists s_{\text{exist}} \in \mathcal{N} \text{ such that } \theta(s_{\text{exist}}, q_i) < \tau \quad (9)$$

Let τ be the roughness threshold and lib be the library size. For the prediction target point s_{target} , the process to search for the optimal set $\mathcal{N}$ of K -nearest neighbors in M_Y is as follows.

(1) Calculate ranking: Compute the state-space distance between each point in the library and $\psi(y, s_{\text{target}})$, and sort the points in ascending order of distance to obtain the candidate queue $\mathcal{Q} = \{q_1, q_2, \dots, q_k\}$.

(2) Iterative screening: Iterate over each candidate point q_i in queue $\mathcal{Q}$, and conduct the target independence test and set independence test in sequence.

(3) Update of K -nearest neighbor set: Only the candidate point that satisfies both constraints is added to the final neighbor set $\mathcal{N}$. The algorithm terminates when $|\mathcal{N}|$ reaches the predefined number of neighbors K or all candidate points have been traversed.

This screening mechanism can filter out points with similar attribute values only because of geographic proximity. It forces the algorithm to find points that may be geographically distant or connected by a rough path, but are truly similar in dynamic state.

Causal Relationship Discrimination

We perform cross-mapping prediction using the screened neighbor set $\mathcal{N} = \{s_{n_1}, \dots, s_{n_K}\}$. The predicted value $\hat{X}(s) | M_Y$ is the weighted average of the X values corresponding to the K -nearest neighbors:

$$\hat{X}(s) | M_Y = \sum_{j=1}^K w_j X(s_{n_j}) \quad (10)$$

$$w_j = \frac{\psi(x, s_i), \psi(x, s_{n_j})}{\exp\left(-\frac{\text{dis}(\psi(x, s_{n_j}), \psi(x, s))}{\text{dis}(\psi(x, s_1), \psi(x, s))}\right)} \quad (11)$$

where the weight w_j is determined based on the state-space distance and its exponential decay. Finally, the Pearson correlation coefficient $\rho_{X \rightarrow Y}$ between the predicted value $\hat{X}$ and the observed value X is computed as the skill score of convergent cross-mapping.

$$\rho_{X \rightarrow Y} = \frac{\text{Cov}(X, \hat{X})}{\sigma_X \cdot \sigma_{\hat{X}}} \quad (12)$$

where $\text{Cov}(X, \hat{X})$ denotes the population covariance, and σ_X and $\sigma_{\hat{X}}$ are the population standard deviations of X and $\hat{X}$, respectively.

To robustly identify the causal direction and strength, this study proposes a testing strategy based on spatial path roughness. If the ρ value is high when τ is small but exhibits a significant and rapid downward trend as τ increases, it indicates that the causal inference mainly depends on spatial autocorrelation, which corresponds to spurious causality or overestimated weak causality. If the ρ value remains relatively stable with increasing τ , or decreases slightly but still maintains a high level at large τ , it indicates that the variables still have strong predictive ability even after removing the effect of spatial autocorrelation, which represents a true causal relationship.

The Computational Complexity of SA-GCCM

Let N denote the total number of spatial units in the study area and E denote the embedding dimension. The computational complexity of each stage is summarized below:

Stage 1: State space reconstruction.

This step iterates through all N spatial units and constructs E -dimensional embedding vectors for each unit based on spatial lags. The time complexity of this stage is $O(N \cdot E)$.

Stage 2: K -nearest neighbor screening.

For each target point, calculating the state-space distance between it and every other point in the library takes $O(N \cdot E)$. The distances are then sorted in ascending order, which takes $O(N \cdot \log N)$. Finally, the spatial path roughness test is applied sequentially to the top C candidate points in the queue. Assuming the average number of sampling points along a spatial path is m , this step takes approximately $O(C \cdot m)$. After iterating over all N target points, the total time complexity of this stage is $O(N \cdot (N \cdot E + N \cdot \log N + C \cdot m))$. When N is large, the time complexity of this stage approximates to $O(N^2 \cdot \log N)$.

Stage 3: Cross-mapping prediction and causal discrimination.

Performing simplex projection with the selected K valid neighbors to generate predictions takes $O(N \cdot K)$. Subsequently, calculating the Pearson correlation coefficient between predicted and observed values takes $O(N)$. Since K is a small constant (e.g., $K = E + 1$ in this study), the time complexity of this stage is $O(N \cdot K)$.

Combining all stages, since $N \gg E$, the overall time complexity of SA-GCCM is dominated by Stage 2, which is $O(N^2 \log N)$.

Experiment and Analysis

To verify the effectiveness of the SA-GCCM method in reducing the interference of spatial autocorrelation and determining the true causal direction, this study selects two representative cases to assess its performance against the GCCM model under different scenarios. The first case explores the causal relationship between nighttime-light data (NL) and topographic factors (elevation and slope), and the second case analyzes the causal relationship between net primary productivity (NPP) and climatic factors (precipitation and temperature).

Simulation Experiment

Simulated Data

To evaluate the performance of SA-GCCM under different spatial autocorrelation strengths and causal effect magnitudes, this study generated synthetic datasets with known causal relationships using the spatial autoregressive model.

We generated variables with varying spatial autocorrelation strengths via Gaussian filtering to replicate the spatial continuity of real geographic data. Gaussian filtering transforms independent and identically distributed random noise into a continuous field with a predefined spatial correlation range via convolution. The spatial autocorrelation strength is controlled by the standard deviation σ . A larger σ corresponds to stronger spatial autocorrelation.

The confounding variable Z is generated as follows:

$$Z_{2d} = \mathcal{G}_\sigma(U_{2d}) \quad (13)$$

where $U_{2d} \in \mathbb{R}^{G \times G}$ is a two-dimensional random matrix drawn from the uniform distribution $\mathcal{U}(0, 1)$, and $\mathcal{G}_\sigma(\cdot)$ denotes the 2D Gaussian filtering operation with standard deviation σ . The treatment variable X is generated with endogeneity, i.e., it is simultaneously influenced by the confounding variable Z and spatial random noise:

$$X = \alpha Z + X_{noise} \quad (14)$$

where $\alpha = 0.05$ is the effect coefficient of Z on X . X_{noise} follows the same generation procedure as Z , and the same σ is used to ensure that X and Z have identical

spatial autocorrelation strengths. The outcome variable Y is generated using a spatial autoregressive model:

$$Y = \rho W_{std} Y + \beta X + \gamma Z + \epsilon \quad (15)$$

where ρ is the spatial autoregressive coefficient controlling the global spatial dependence of Y , fixed at $\rho = 0.4$; β is the direct causal effect of X on Y ; γ is the effect coefficient of Z on Y , fixed at $\gamma = 2$; and ϵ is an independent and identically distributed random error term drawn from the normal distribution $N(0, 1)$.

By adjusting the direct causal effect β of X on Y and the Gaussian filter bandwidth σ , we generate synthetic datasets with varying spatial autocorrelation levels and causal strengths to validate the performance of SA-GCCM across different scenarios.

Analysis of Simulation Experiment Results

Nine parameter combinations were tested in this study (true causal effect $\beta = 1, 1.5, 2$; spatial autocorrelation strength $\sigma = 6, 8, 10$). We evaluated the performance of SA-GCCM based on $\rho - \tau$ curve characteristics and compared it with two baseline methods: Pearson correlation coefficient and GeoDetector.

We calculated the cross-mapping prediction results between X and Y across different roughness thresholds (see Supplementary Fig. S1 for details) and extracted the optimal predictive performance at each roughness level. The $\rho - \tau$ curves for different spatial autocorrelation strengths and causal effects are presented in Fig. 3. In all nine experiments, the maximum ρ of the true causal direction $X \rightarrow Y$ (red curves) decreased slowly and remained stable as τ increased, while the maximum ρ of the reverse spurious causal direction $Y \rightarrow X$ (blue curves) decreased rapidly and significantly. This confirms that the proposed method can effectively remove spurious causal signals induced by spatial autocorrelation using the spatial path roughness threshold.

Table 1 presents a comparison of the results of SA-GCCM, Pearson correlation coefficient, and GeoDetector across different parameter combinations. Specifically, the ρ value for SA-GCCM was defined as the stable value at $\tau = 0.5$ (when spurious causal signals had been sufficiently removed). The Pearson correlation coefficient measures the linear correlation between variables, and the q value of GeoDetector measures the explanatory power of the independent variable for the dependent variable. Compared with Pearson correlation coefficient and GeoDetector, SA-GCCM not only can identify causal directions but also quantify causal effect magnitudes more accurately,

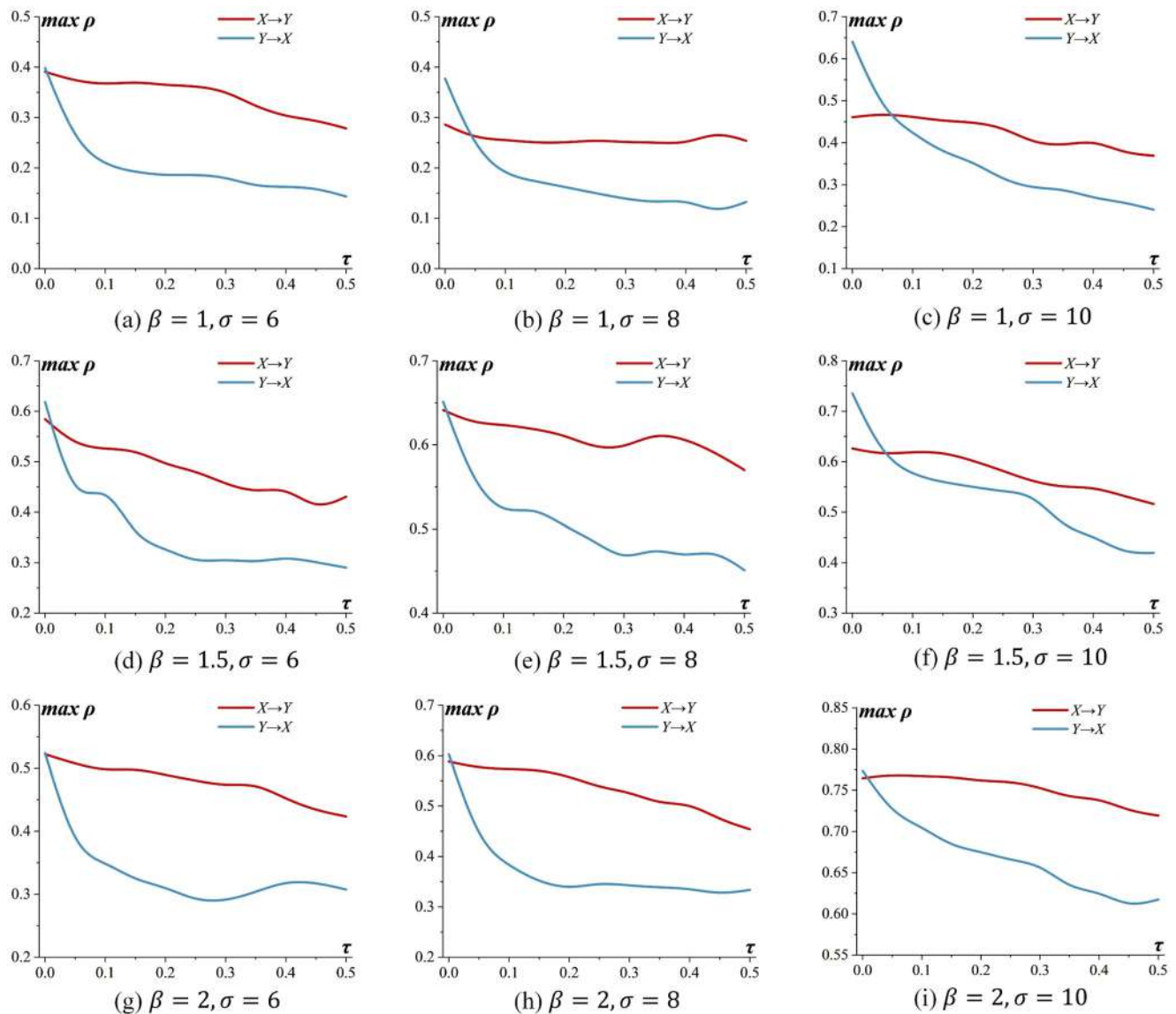


Fig. 3 $\rho - \tau$ curves under different spatial autocorrelation strengths and causal effects. (a) - (i) correspond to the parameter combinations $(\beta = 1, \sigma = 6)$, $(\beta = 1, \sigma = 8)$, $(\beta = 1, \sigma = 10)$, $(\beta = 1.5, \sigma = 6)$, $(\beta = 1.5, \sigma = 8)$, $(\beta = 1.5, \sigma = 10)$,

$(\beta = 2, \sigma = 6)$, $(\beta = 2, \sigma = 8)$ and $(\beta = 2, \sigma = 10)$, respectively. Causal inference under different spatial autocorrelation strengths and causal effects

which gives it significant advantages in spatial causal inference tasks.

The Causality Between Nighttime-light Intensity and Topographic Factors

Topography is a key factor that affects population density and industrial development, and nighttime-light data acts as an important indicator of socioeconomic development and energy consumption. Generally, areas with flat terrain and low elevation are more suitable for human settlement and industrial development.

In this experiment, the Beijing-Tianjin-Hebei region and its surrounding areas were selected as the study area (115.5°E–117.5°E, 39.25°N–41.25°N) to analyze the causal relationships among elevation, slope and nighttime-light. The spatial distribution of these datasets is presented in Fig. 4. The elevation data used in this study were derived from the SRTM digital elevation model provided by the CGIAR-CSI website (<http://srtm.csi.cgiar.org/>). Based on the elevation raster data, the elevation change rates of the central raster cell in different directions were calculated using the difference method, and the slope data were further generated from these calculations. The nighttime-light data

Table 1 Comparison of results from different methods under different spatial autocorrelation strengths and causal effects

Parameter	Pearson correlation coefficient	GeoDetector	SA-GCCM
$\beta = 1, \sigma = 6$	0.487	0.220	$\rho_{X \rightarrow Y} = 0.278$ $\rho_{Y \rightarrow X} = 0.143$
$\beta = 1, \sigma = 8$	0.328	0.107	$\rho_{X \rightarrow Y} = 0.253$ $\rho_{Y \rightarrow X} = 0.132$
$\beta = 1, \sigma = 10$	0.489	0.233	$\rho_{X \rightarrow Y} = 0.369$ $\rho_{Y \rightarrow X} = 0.240$
$\beta = 1.5, \sigma = 6$	0.553	0.274	$\rho_{X \rightarrow Y} = 0.430$ $\rho_{Y \rightarrow X} = 0.290$
$\beta = 1.5, \sigma = 8$	0.644	0.395	$\rho_{X \rightarrow Y} = 0.570$ $\rho_{Y \rightarrow X} = 0.451$
$\beta = 1.5, \sigma = 10$	0.623	0.398	$\rho_{X \rightarrow Y} = 0.516$ $\rho_{Y \rightarrow X} = 0.419$
$\beta = 2, \sigma = 6$	0.608	0.299	$\rho_{X \rightarrow Y} = 0.423$ $\rho_{Y \rightarrow X} = 0.307$
$\beta = 2, \sigma = 8$	0.643	0.348	$\rho_{X \rightarrow Y} = 0.45$ $\rho_{Y \rightarrow X} = 0.333$
$\beta = 2, \sigma = 10$	0.776	0.563	$\rho_{X \rightarrow Y} = 0.719$ $\rho_{Y \rightarrow X} = 0.617$

were obtained from the Long-term Annual Artificial Nighttime-Light Dataset of China (Zhang et al. 2024). All datasets were resampled to a consistent spatial resolution of 0.01° (approximately 1.11 km), and all data correspond to the year 2020. A summary of these datasets is provided in Table 2.

There is a unidirectional causal relationship between nighttime-light intensity and topographic factors; that is, topographic factors drive nighttime-light intensity (Zhang et al. 2019; Huang et al. 2023). Figure 5 presents the causal inference results of GCCM and SA-GCCM, with the embedding dimension E set to 3 and the number of nearest neighbors set as $K = E + 1$. For the relationship between elevation and nighttime-light intensity, GCCM achieves high predictive performance in both directions, and both curves show strong convergence (Fig. 5(a)). If we only use the convergence criterion of GCCM, we may reach an incorrect conclusion that “changes in nighttime-light intensity

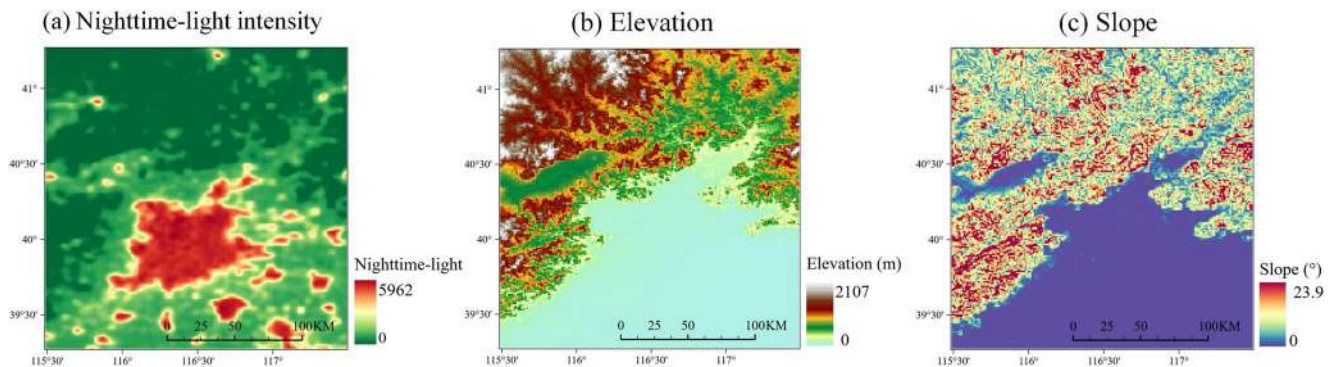
Table 2 Summary of nighttime-light data and its influencing factors

Data	Variable	Spatial resolution	Unit
Nighttime-light data	NL	0.01°	-
Elevation data	Elev	30 m	m
Slope data	Slope	30 m	$^\circ$

cause variations in elevation. For the relationship between slope and nighttime-light intensity, causal misidentification also occurs (Fig. 5(b)). The predictive skill of $NL \rightarrow Slope$ (blue line) is significantly higher than that of $Slope \rightarrow NL$ (red line). Therefore, when the spatial autocorrelation of variables is ignored, the causal inference results obtained by the GCCM algorithm are misleading.

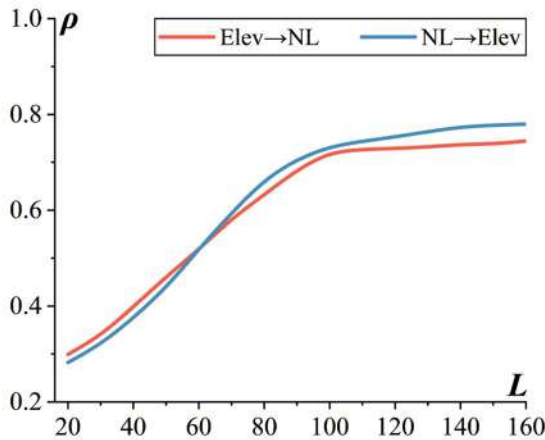
We calculated the cross-mapping prediction results between nighttime-light intensity and topographic factors under different roughness thresholds (see Supplementary Figs. S2 and S3 for details), and extracted the optimal cross-prediction performance at each roughness level. The $\rho - \tau$ curves were then plotted (Fig. 5(c) and (d)). The results show that after incorporating the spatial path roughness constraint, the SA-GCCM method can effectively identify the true causal direction. As the roughness threshold τ increases, the predictive performance of the true causal directions ($Elev \rightarrow NL$, $Slope \rightarrow NL$) remains highly robust. In contrast, the predictive performance of the reverse causal relationships ($NL \rightarrow Elev$, $NL \rightarrow Slope$) decreases significantly with increasing τ . In Fig. 5(c), the predictive skill of $NL \rightarrow Elev$ drops rapidly from 0.79 to 0.3; in Fig. 5(d), the predictive skill of $NL \rightarrow Slope$ decreases sharply from 0.6 to 0.26. These results confirm that the reverse causal signals identified by GCCM partly stem from the interference of spatial autocorrelation. Once the “false neighbors” induced by spatial autocorrelation are removed, the accuracy of predicting nighttime-light intensity using topographic factors decreases significantly.

To verify whether the SA-GCCM method can effectively remove “false neighbors” caused by spatial autocorrelation, we further analyzed the geographic distance distribution

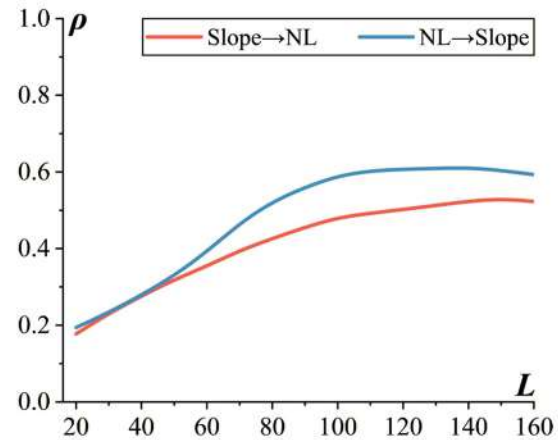
**Fig. 4** Spatial patterns of nighttime-light data and topographic influencing factors in the study area. **a** shows the spatial distribution of nighttime-light intensity, reflecting human activity concentration. **b**

shows the elevation data. **c** shows the slope data. Spatial distribution of nighttime-light data and its influencing factors

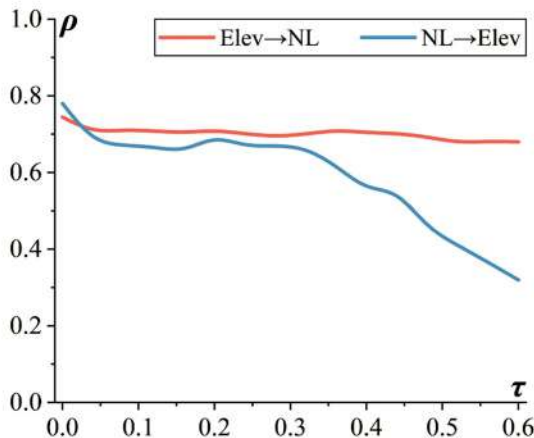
(a) Causal inference between Elev and NL (GCCM)



(b) Causal inference between Slope and NL (GCCM)



(c) Causal inference between Elev and NL (SA-GCCM)



(d) Causal inference between Slope and NL (SA-GCCM)

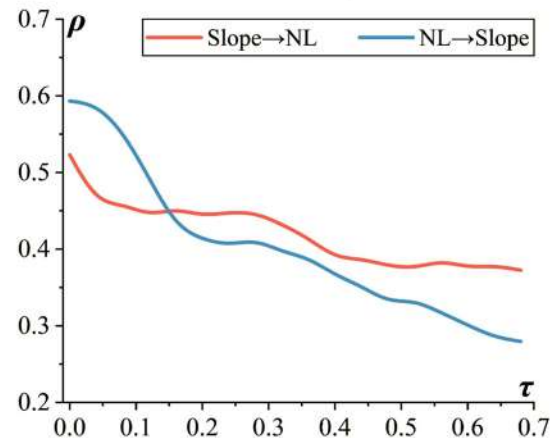


Fig. 5 **a** shows the causal inference results between Elev and NL derived from GCCM; **b** presents the causal inference results between Slope and NL derived from GCCM; **c** demonstrates the causal inference results between Elev and NL derived from SA-GCCM; **d** illus-

trates the causal inference results between Slope and NL derived from SA-GCCM. Causal inference between nighttime-light and topographic factors

between the target point and its K ($K = 4$) nearest neighbors used for prediction (Fig. 6). By comparing the geographic distances between the target point and its K -nearest neighbors at $\tau = 0$ and $\tau = 0.4$, we find that the roughness threshold can effectively remove “false neighbors” induced by spatial autocorrelation. After introducing the path roughness constraint, the average spatial distance between the target point and its K -nearest neighbors increases significantly. This result indicates that SA-GCCM skips spatially adjacent neighbors and instead searches for points with similar dynamic states across a broader spatial range.

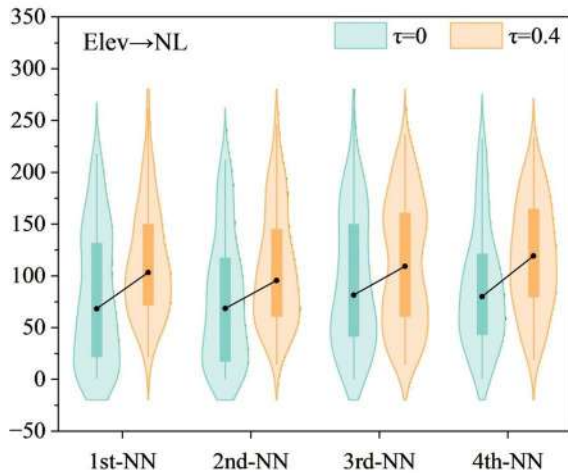
This mechanism fundamentally prevents GCCM from degenerating into “local inverse distance weighted interpolation” and ensures that causal inference relies on long-range dynamic similarity instead of geographic proximity. Combined with Fig. 5(c) and Fig. 5(d), although true causal directions use more distant neighbors for prediction, their

predictive skill ρ remains stable. In contrast, reverse causal directions tend to select spatially close points, but their predictive ability decreases sharply under a large τ . This comparison clearly demonstrates that real causal coupling is embedded in the global dynamic structure of the system and enables reliable prediction even across large geographic distances. However, spurious causality depends strongly on local spatial structures; once “false neighbors” induced by spatial autocorrelation are removed, its predictive performance decreases rapidly.

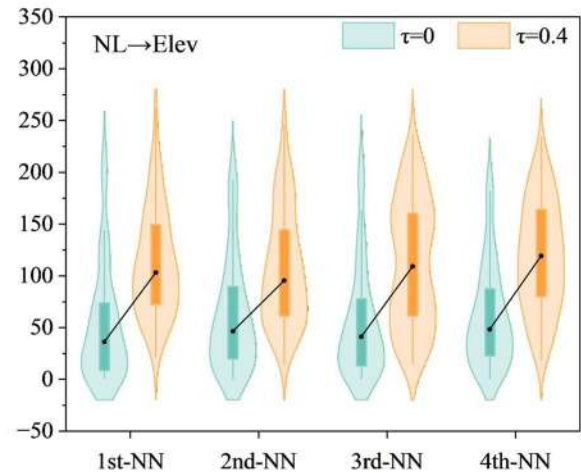
The Causality Between Net Primary Productivity and Climate Factors

As the key climatic factors controlling plant photosynthetic efficiency, growing season length, and nutrient supply, temperature and precipitation directly determine the spatial

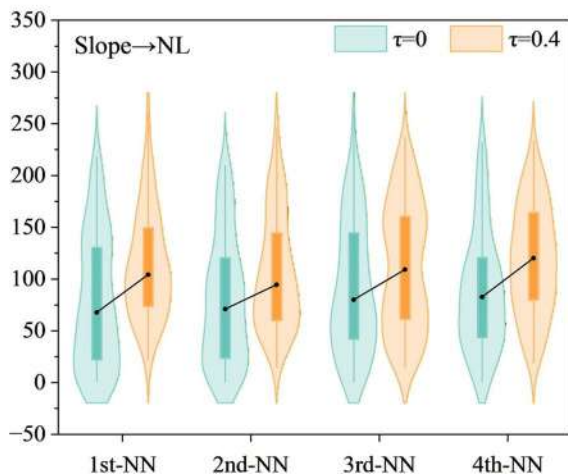
(a) Comparison of spatial distances (Elev→NL)



(b) Comparison of spatial distances (NL→Elev)



(c) Comparison of spatial distances (Slope→NL)



(d) Comparison of spatial distances (NL→Slope)

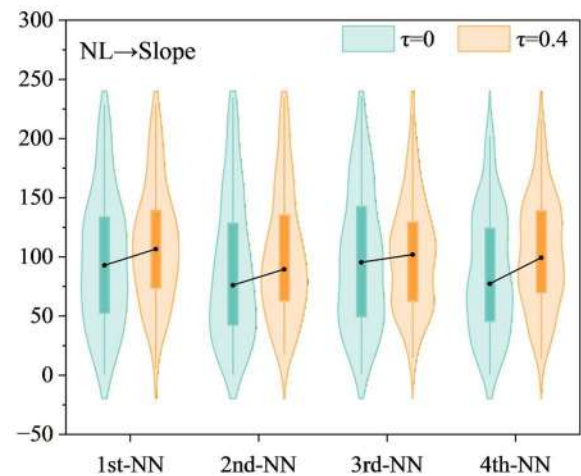


Fig. 6 **a** shows the spatial distances between K -nearest neighbors and target points for the Elev → NL causal direction under different roughness thresholds ($\tau = 0$ and $\tau = 0.4$); **b** presents the spatial distances for the NL → Elev causal direction; **c** demonstrates the spatial distances for the Slope → NL causal direction; **d** illustrates

the spatial distances for the NL → Slope causal direction. Comparison of spatial distances between K -nearest neighbors and target points under different roughness thresholds (nighttime-light intensity and topographic factors case)

distribution pattern of NPP. In contrast, NPP only has a weak impact on local microclimates. In local ecosystems, the causal effect of temperature and precipitation on NPP is significant and dominant, with a strength greater than the feedback effect of NPP on climate (Jia et al. 2024; Liu et al. 2026).

This study focuses on a region in southwestern China (103.5°E–105.5°E, 27.25°N–29.25°N) to analyze the causal relationships between NPP, precipitation, and temperature. This region has complex terrain and high spatial heterogeneity of hydrothermal conditions, making it ideal for testing causal inference methods. The spatial distribution of these datasets is presented in Fig. 7. The NPP data used in this study were derived from the NASA Earth Science Data

website (<https://lpdaac.usgs.gov>). Temperature and precipitation data were obtained from the China Meteorological Forcing Dataset (He et al. 2020; Jiang et al. 2023). All datasets were resampled to a spatial resolution of 0.01° and corresponded to the year 2024. A summary of these datasets is provided in Table 3.

Similarly, we calculated the cross-mapping prediction results between NPP and climatic factors under different roughness thresholds (see Supplementary Figs. S4 and S5 for details), and plotted the $\rho - \tau$ curves. For the relationship between precipitation and NPP, GCCM shows that the predictive skill of NPP → Pre is higher than that of Pre → NPP (Fig. 8(a)). However, SA-GCCM indicates that the predictive skill of NPP → Pre decreases rapidly

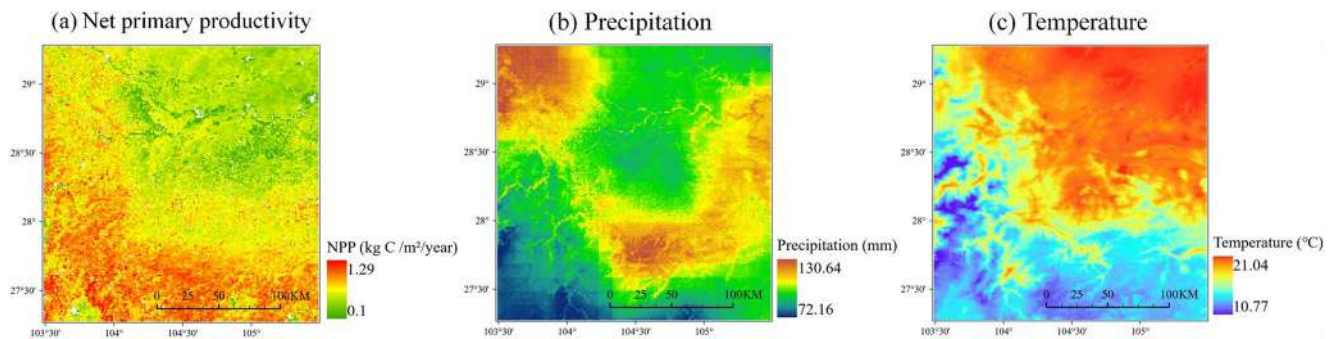
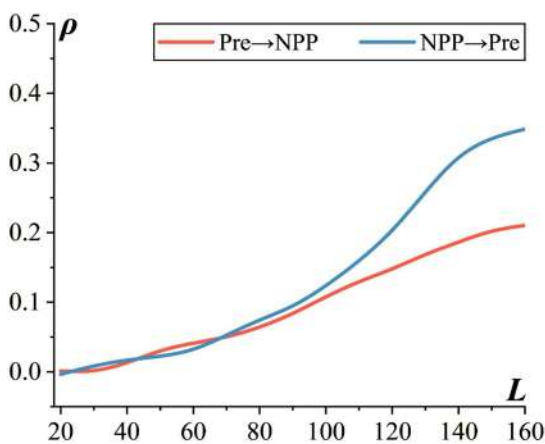


Fig. 7 **a** shows the spatial distribution of NPP. **b** shows the precipitation data. **c** shows the temperature data. Spatial distribution of NPP data and its influencing factors

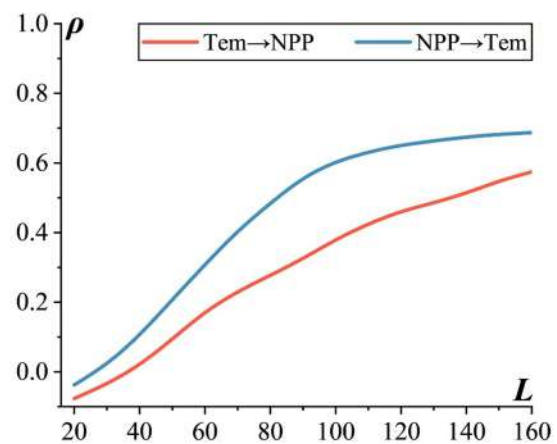
Table 3 Summary of NPP and its influencing factors

Data	Variable	Spatial resolution	Unit
Net primary productivity	NPP	0.01 °	kg C/m ² /year
Precipitation data	Pre	0.01 °	mm
Temperature data	Tem	0.01 °	°C

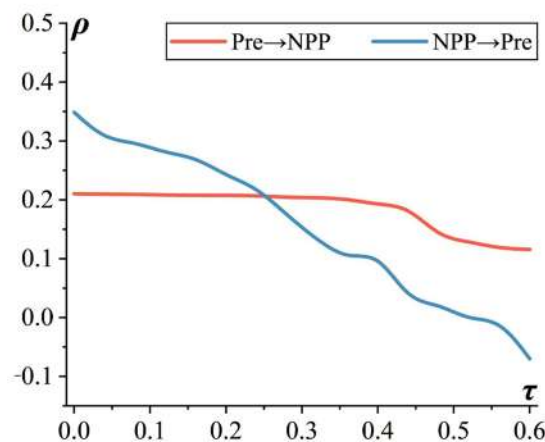
(a) Causal inference between Pre and NPP (GCCM)



(b) Causal inference between Tem and NPP (GCCM)



(c) Causal inference between Pre and NPP (SA-GCCM)



(d) Causal inference between Tem and NPP (SA-GCCM)

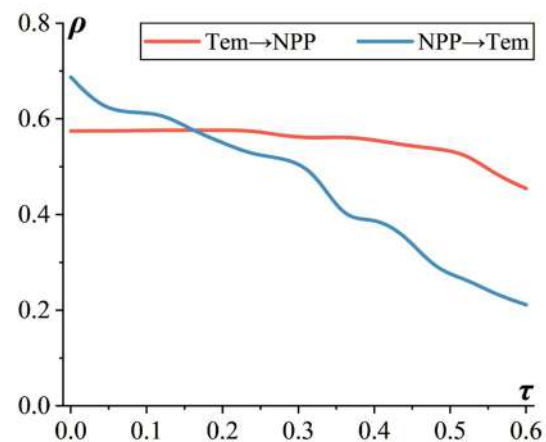


Fig. 8 **a** shows the causal inference results between precipitation and NPP derived from GCCM; **b** presents the causal inference results between temperature and NPP derived from GCCM; **c** demonstrates the causal inference results between precipitation and NPP derived

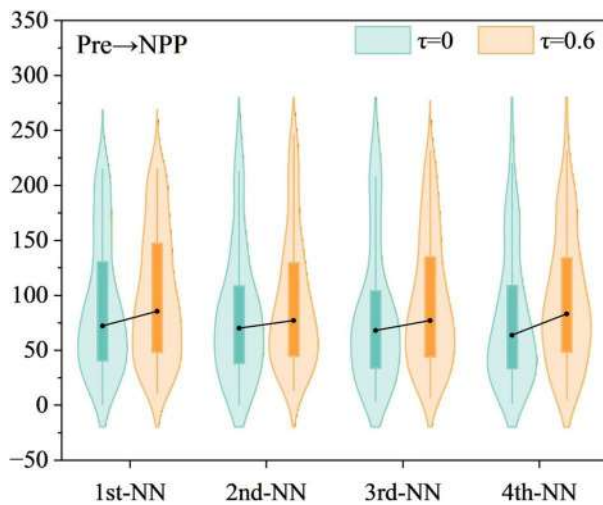
from SA-GCCM; **d** illustrates the causal inference results between temperature and NPP derived from SA-GCCM. Causal inference between NPP and climate factors

with increasing τ (Fig. 8(c)). The predictive performance of $\text{NPP} \rightarrow \text{Pre}$ is highly sensitive to τ , which suggests that its high predictive skill mainly stems from spatial autocorrelation. In contrast, although the average predictive skill of $\text{Pre} \rightarrow \text{NPP}$ (red line) is relatively low (approximately 0.2), the p value remains stable as τ increases. For the relationship between temperature and NPP, the GCCM results show that NPP exerts a stronger causal effect on temperature (Fig. 8(b)). In contrast, SA-GCCM shows that $\text{Tem} \rightarrow \text{NPP}$ is highly stable (Fig. 8(d)). This indicates that temperature is the real driver of NPP changes, and its predictive performance does not drop significantly

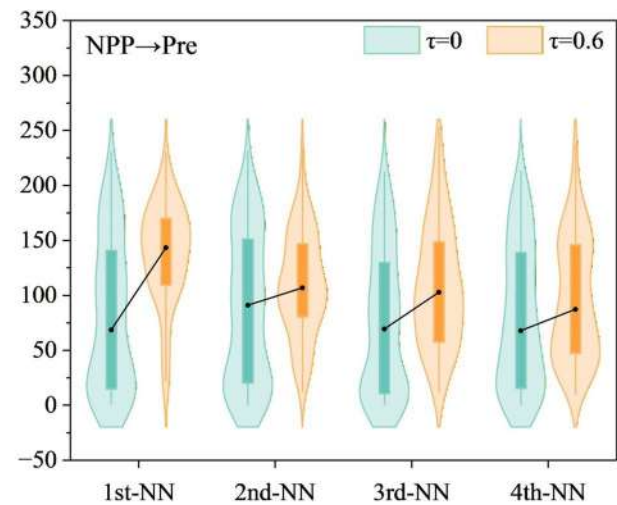
as τ increases. In contrast, the predictive performance of $\text{NPP} \rightarrow \text{Tem}$ decreases sharply to approximately 0.2, showing that the feedback effect of NPP on temperature is weak. This result agrees with previous studies.

Similarly, we analyzed the geographic distance distribution between target points and their K -nearest neighbors at $\tau = 0.6$ (Fig. 9). The comparison demonstrates that the roughness threshold can also effectively reduce local spatial autocorrelation. Statistics show that the average spatial distance between target points and their K -nearest neighbors increases substantially after adding the roughness constraint. This implies that SA-GCCM successfully forces

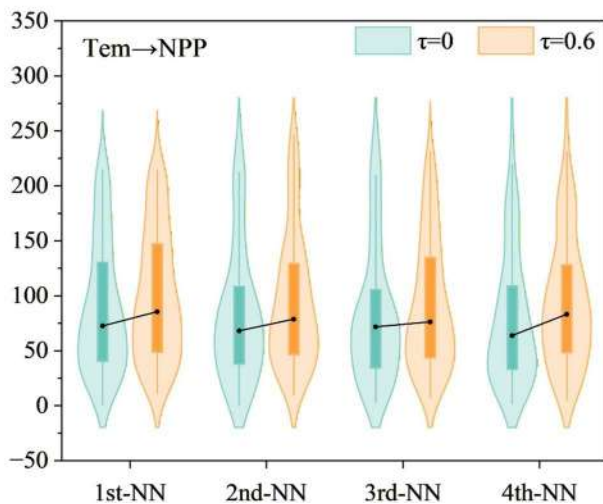
(a) Comparison of spatial distances ($\text{Pre} \rightarrow \text{NPP}$)



(b) Comparison of spatial distances ($\text{NPP} \rightarrow \text{Pre}$)



(c) Comparison of spatial distances ($\text{Tem} \rightarrow \text{NPP}$)



(d) Comparison of spatial distances ($\text{NPP} \rightarrow \text{Tem}$)

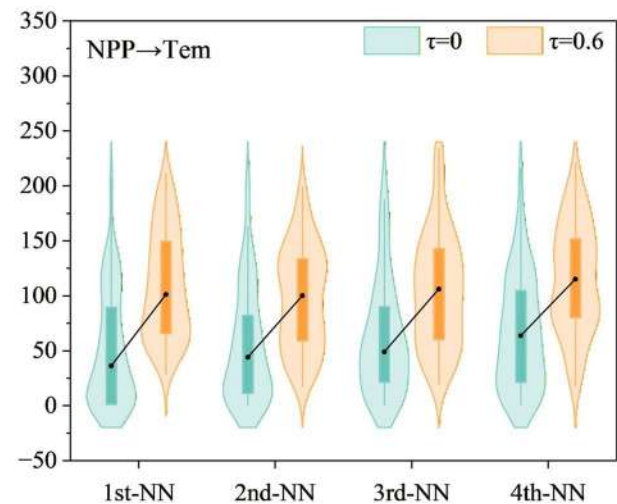


Fig. 9 **a** shows the spatial distances between K -nearest neighbors and target points for the $\text{Pre} \rightarrow \text{NPP}$ causal direction under different roughness thresholds ($\tau = 0$ and $\tau = 0.6$); **b** presents the spatial distances for the $\text{NPP} \rightarrow \text{Pre}$ causal direction; **c** demonstrates the

spatial distances for the $\text{Tem} \rightarrow \text{NPP}$ causal direction; **d** illustrates the spatial distances for the $\text{NPP} \rightarrow \text{Tem}$ causal direction. Comparison of spatial distances between K -nearest neighbors and target points under different roughness thresholds (NPP and climatic factors case)

the algorithm to skip geographically close neighbors, and instead search for points with similar dynamic states within a larger spatial range.

Discussion

Parameter Sensitivity Analysis

SA-GCCM has two key parameters: the embedding dimension E and the spatial path roughness threshold τ . The following will discuss the impact of these two parameters on the SA-GCCM model.

The Influence of Embedding Dimension E on SA-GCCM Model

The embedding dimension E determines whether the reconstructed state space has a complete topological structure. According to Takens' embedding theorem, E should be sufficiently large to unfold the dynamical attractor of the system, thus avoiding trajectory overlap caused by overly small E . However, an excessively large E adds redundant information. In SA-GCCM, the selection of E follows the same rules as CCM. Its value can be determined from the optimal prediction results of simplex projection. To verify the sensitivity of the SA-GCCM algorithm to the parameter τ , we compared the $\rho - \tau$ curves under different E values. As shown in the Fig. 10, SA-GCCM is not sensitive to E , as the curves show only small differences across different E .

The Influence of Spatial Path Roughness on SA-GCCM Model

The spatial path roughness threshold τ is the key parameter of the SA-GCCM. When $\tau = 0$, SA-GCCM reduces to GCCM and cannot effectively remove "false neighbors" induced by spatial autocorrelation. Conversely, if the value of τ is too large, the screening conditions become excessively strict. This causes a sharp drop in the number of valid neighbors and may even damage the local manifold structure, making simplex projection invalid because of insufficient reference samples. Our results show that the robustness of SA-GCCM does not depend on a fixed τ , but on how ρ changes with τ . For real causal relationships (e.g., $Tem \rightarrow NPP$, $Elev \rightarrow NL$), ρ remains stable over a wide range as τ increases. In this range, even when spatially autocorrelated neighbors are removed, predictive accuracy stays high or decreases only slowly. This indicates that the causal relationships between variables within the system are intrinsic and do not depend on spatial smoothness.

For spurious or weak causal relationships (e.g., $NPP \rightarrow Pre$, $NL \rightarrow Slope$), ρ is very sensitive to τ and decreases quickly as τ increases. Therefore, in practical applications, it is not recommended to set τ to a fixed value. Instead, a parameter sweeping strategy should be adopted to diagnose the nature of the causal relationship by plotting the $\rho - \tau$ curves. This dynamic analysis is the main advantage of SA-GCCM over traditional statistical methods. Future research could explore adaptive τ value selection algorithms, such as automatically determining a reasonable roughness threshold range based on the statistical characteristics of the study area, thereby further reducing the subjectivity of manual parameter setting.

To avoid blind parameter tuning in practical applications, we established the prior range of τ using quantiles of the spatial path roughness distribution. Specifically, we selected the 25th and 75th percentiles of the roughness distribution for state-similar point pairs as this prior range. The 25th percentile is the lower quartile, representing the threshold below which 25% of the roughness values fall when all state-similar point pairs are sorted in ascending order; equivalently, 75% of pairs have roughness values above this threshold. This range is directly derived from the intrinsic statistical distribution of the data and can automatically adapt to the spatial characteristics of different study areas.

Differences Between SA-GCCM and Spatial Interpolation Methods

Co-kriging, multivariate smoothing, and multivariate splines are widely used classical geospatial methods with excellent performance in spatial interpolation and spatial prediction. However, these methods differ fundamentally from SA-GCCM. These interpolation methods aim to improve prediction accuracy for unknown points by exploiting spatial autocorrelation between variables. In contrast, SA-GCCM is a causal inference method whose core objective is to uncover nonlinear driving mechanisms and causal directions between variables. Although SA-GCCM involves cross-mapping prediction, improving prediction accuracy is not its primary objective. When a causal relationship exists between two variables, the Pearson correlation coefficient between the predicted value $\hat{X}$ and the observed value X increases and converges to a stable value as the library size grows. Conversely, if no causal relationship exists, improving the cross-mapping process will not yield better prediction results.

Spatial Non-stationarity and Local Spatial Causal Inference

Spatial non-stationarity is an inherent property of geographic systems and a critical challenge for all global spatial

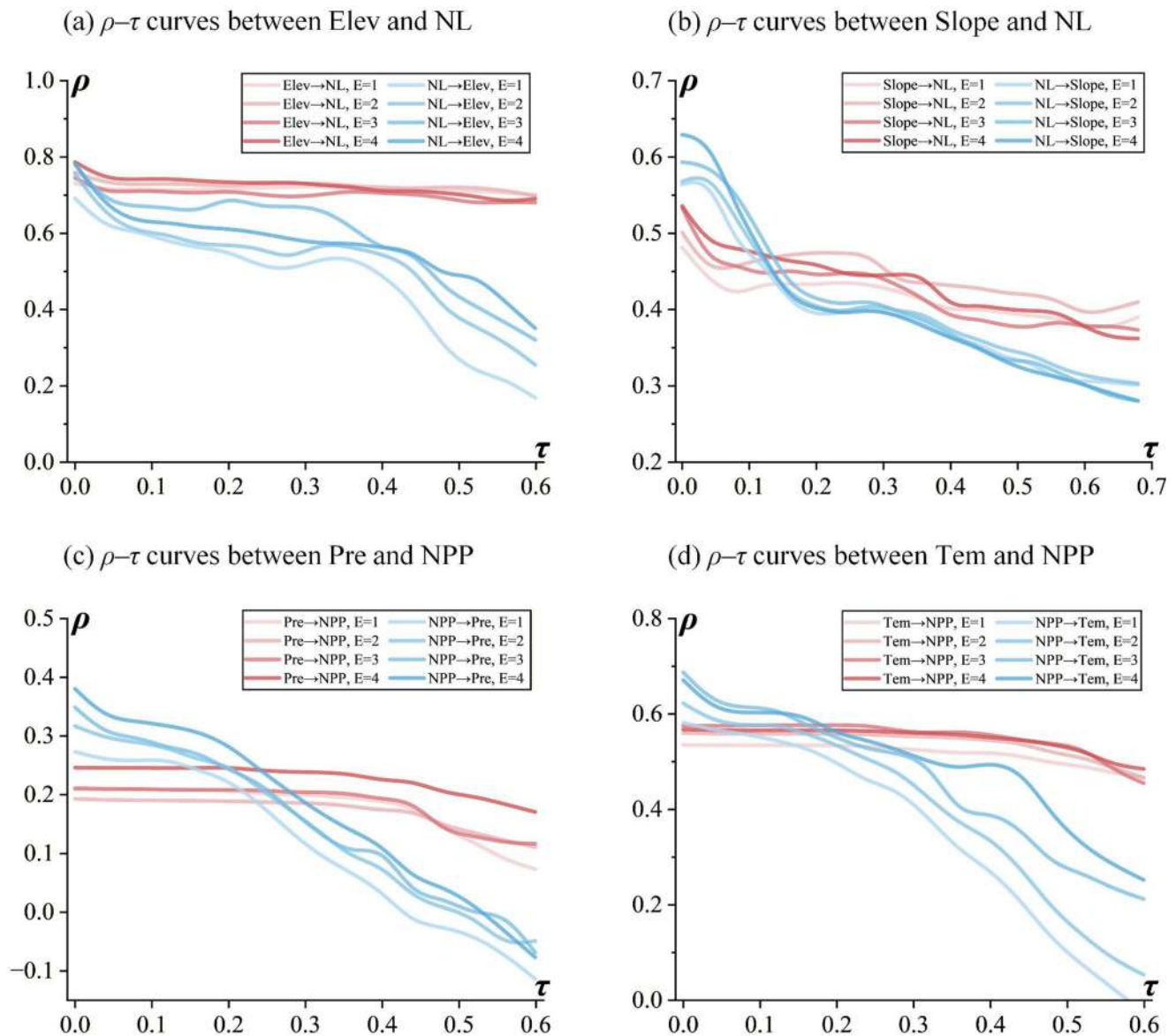


Fig. 10 **a** shows the ρ - τ curves for the causal relationship between Elev and NL under different embedding dimensions ($E=1, 2, 3, 4$); **b** presents the ρ - τ curves for the causal relationship between Slope and NL under different embedding dimensions; **c** demonstrates the

ρ - τ curves for the causal relationship between Pre and NPP under different embedding dimensions; **d** illustrates the ρ - τ curves for the causal relationship between Tem and NPP under different embedding dimensions. ρ - τ curves under different embedding dimensions

causal inference methods. Like the original GCCM, the proposed SA-GCCM presupposes a unified causal dynamic mechanism throughout the study area. This global stationarity assumption enables SA-GCCM to identify the dominant causal direction for the entire study area, but it also implies that SA-GCCM's outputs essentially represent a weighted average of causal effects across different sub-regions, thereby masking the true local variability in causal strength.

To address the spatial non-stationarity limitation of global models, Geographically Weighted Regression (Brunsdon et al. 1998) provides a classic solution: it constructs local neighborhoods with adaptive bandwidths for each spatial

unit, generating spatially heterogeneous regression coefficients. However, this window-based approach is not suitable for manifold-based spatial causal inference, for two key reasons:

First, confining analysis to local neighborhoods reintroduces spatial autocorrelation interference, which contradicts the core design objective of SA-GCCM. SA-GCCM is specifically designed to eliminate false neighbors induced by spatial autocorrelation using the spatial path roughness index, enabling the algorithm to search for state-similar point pairs across the entire study area. When the analysis is restricted to a local region, variable relationships within

the neighborhood are often dominated by spatial smoothing, leading to strong linear correlations. In this scenario, the local ρ value will approach 1, and SA-GCCM will effectively degenerate into a local spatial interpolation technique.

Second, sliding windows do not align with natural geographic boundaries, and parameter selection lacks objective criteria. Geographic processes typically exhibit irregular, patchy spatial boundaries; thus, regular sliding windows or continuous kernel functions inevitably cross regions with distinct driving mechanisms, confounding heterogeneous causal processes. Additionally, the selection of window size or bandwidth relies heavily on subjective experience, and different parameter settings lead to inconsistent distributions of local causal strengths.

Therefore, partitioning the study area into homogeneous sub-regions based on prior geographic knowledge is a more practical approach for local causal inference using SA-GCCM. The core idea is to divide the study area into sub-regions with relatively consistent internal driving mechanisms, run SA-GCCM independently in each sub-region, and reveal the spatial heterogeneity of causal relationships by comparing the characteristics of $\rho - \tau$ curves and the magnitudes of causal effects across different sub-regions.

Limitations and Future Work

Although SA-GCCM performs well in reducing spatial autocorrelation interference, it has two main limitations. First, existing studies only consider bivariate causal inference. In complex Earth systems, multiple factors jointly influence a dependent variable. Future research could attempt to extend SA-GCCM to a multivariate causal inference framework using multivariate embedding. Furthermore, the core of SA-GCCM lies in calculating the spatial path roughness between two points and performing iterative screening. For large-scale, high-resolution raster data (such as global-scale remote sensing imagery), its computational cost grows exponentially. Future work should adopt spatial indexing or parallel computing to improve the computational efficiency of SA-GCCM for big spatial data.

Conclusion

To solve the problem that GCCM is susceptible to interference from “false neighbors” and thus yields biased causal inference, this study proposes a Geographical Convergent Cross Mapping accounting for Spatial Autocorrelation. By introducing the spatial path roughness to quantify the smoothness of attribute changes between two points in geographic space, this method effectively eliminates “false neighbors” caused by spatial autocorrelation. Meanwhile, a

causal identification strategy based on spatial path roughness is proposed, which distinguishes real causal relationships from spurious correlations by analyzing how cross-prediction accuracy changes with spatial path roughness.

Verified by simulation experiment and two typical real cases, SA-GCCM outperforms GCCM in causal identification and can effectively distinguish real dynamic coupling from spurious correlations induced by spatial autocorrelation. In the case of nighttime-light intensity and topographic factors, SA-GCCM confirms the unidirectional driving effect of elevation and slope on nighttime-light intensity. In the case of NPP and climatic factors, it accurately identifies the dominant driving effect of temperature and precipitation on NPP and the weak feedback effect of NPP on these climatic factors.

This study reveals the interference mechanism of spatial autocorrelation on causal inference for spatial cross-sectional data, clarifies the core principle that “false neighbors” in GCCM lead to distortion of the manifold topological structure and the degeneration of dynamic prediction into spatial interpolation. Meanwhile, SA-GCCM provides a tool for geography, ecology, climatology, environmental science, and related disciplines to uncover nonlinear driving mechanisms of complex geographic phenomena and identify real causal relationships between variables. It also provides new insights for future research on multivariate causal inference and causal mechanisms under spatial non-stationarity.

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Author Contributions K.Z. and X.H. conceptualized the study and developed the methodology. K.Z. and S.Y. performed the data analysis, investigation, validation. Y.S. and L.Y. reviewed and edited the manuscript. Y.S. supervised the project. All authors reviewed the manuscript.

Data Availability Data and code is provided within the manuscript.

Declarations

Ethical Approval All authors certify that the research subject of this manuscript does not raise any ethical concerns.

Informed Consent All authors certify that no informed consent was required for the subject matter or materials discussed in this manuscript.

Competing Interests The authors declare no competing interests.

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